

P100-027: KTH boiling two-phase water flow

Principia-100 | Standardized portfolio | 1 October 2026

Scenario and evaluation contract

Domain: thermal-fluid engineering. The current default task predicts Measured local vapor void fraction in dimensionless. Independent unit: complete experimental run. Its packaged cohort contains 36 assigned rows in 3 groups.

All supplied outcomes are now exposed. Historical reserved evaluations remain part of the evidence record; future scores are retrospective. No newly established fundamental law or measured deployment impact is admitted by this packaging.

Current portfolio findings

Retrospective extension: Joint drift-flux/spatial calibration improves local development prediction but cannot identify universal slip constants.

Limit: $C_0=0.87924253$ lies below one and compensates with odds/spatial terms; source coordinate geometry remains unresolved. The later positive-slip revision loses; the joint model remains an empirical calibrated reference.

Unsupported or falsified: A positive-slip physical revision does not improve the prior joint profile; the new flexible control sets a stronger numerical target.

Limit: Earlier joint $C_0=0.879$ can compensate with spatial/intercept terms. Enforcing positive slip alone loses; parameters are not uniquely identified.

Versioned tasks

Task family	Target and information	Source scope
original	Measured local vapor void fraction (dimensionless)	original corpus
round2	Measured local vapor void fraction (dimensionless)	original corpus

Different targets, calibration budgets and cohorts have different task IDs. Their raw error values cannot be pooled into a scenario ranking. Exact equations, coefficients, permitted variables, timing, groups and per-model results are bound in the task packages.

Evaluation: the shared CLI provides numerical scoring, explicit coverage and abstention, uncertainty and event diagnostics. Agent review separately assesses scientific support; no aggregate discovery score is produced.

Detailed reference note: Boiling flow: a conditional radial void-profile reference

Principia-100 · Case 27 · Calibrated within-facility extension

1. Scenario and available measurements

The KTH/Westinghouse HWAT corpus contains 12 boiling-water runs from 2023, released in 2025. Each XML run supplies boundary metadata, axial pressure/wall temperature and 12 radial void, velocity and phase-size records. These are processed experimental products. The target is local vapor void fraction, dimensionless; measured velocity and phase scales are excluded from predictors.

2. Experimental method

Metadata hashing reserved three whole runs and 36 positions; nine runs supported whole-run development validation. Five hypotheses tested slip, spatial profiles, mass-flux dependence, quality/profile interactions and established drift flux. Selection and stopping preceded confirmation. Flexible comparators used identical inputs, with nested training-group folds selecting kernel width and ridge. No target-derived regime label is used. Densities follow fixed IAPWS saturation calculations from absolute pressure; conclusions condition on supplied exit quality.

3. Equation and physical interpretation

Let x denote exit quality, ρ_l and ρ_v saturated densities, and s the native coordinate divided by its maximum sampled value in the run. Define

$$\alpha_h = \frac{x/\rho_v}{x/\rho_v + (1-x)/\rho_l}.$$

$$\text{logit}(\hat{\alpha}) = \text{logit}(\alpha_h) + b_0 + b_2 s^2.$$

Here $b_0 = 0.644498$ and $b_2 = -2.411380$, both dimensionless. The bounded equation describes spatial redistribution of phase odds. It is not an area-averaged conservation closure. Coordinate units are uncertain; s does not establish physical tube-radius or wall-distance geometry. The drift-flux comparator uses $\alpha = j_g / [C_0(j_g + j_l) + V_d]$, with superficial fluxes and drift velocity in m/s. Exact parameters are in `rules.json`.

4. Findings and predictive performance

Primary MAE averages the 12 positions within each run, then the three run errors equally. All 36 assigned positions were scored.

Frozen representation	Mean run MAE (void fraction)
Quadratic profile correction	0.106701
Homogeneous saturated mixture	0.182569
Established drift-flux competitor	0.139195
Constant development mean	0.323480
Fixed flexible RBF	0.207382
Group-tuned flexible RBF	0.254019

The correction lowers aggregate error by 41.56% versus homogeneous and improves all three runs against it. Individual MAE ranges 0.044966-0.187872. Drift flux wins the low-quality Run 32 counterexample: 0.051245 versus 0.187872 for the profile model. Added quality-dependent gradient terms failed development improvement. These are scoped predictive and negative findings; radial redistribution is established prior art and no novel physics is admitted.

5. Practical value and limits

The compact correction supplies an interpretable profile benchmark for thermal-hydraulic code development in this facility. Low quality/pressure confirmation conditions extend beyond development ranges; Run 32 demonstrates weak extrapolation. Source XOUT generation is not fully documented, so boundary conditioning is explicit. Three runs with possible shared-date calibration cannot establish population confidence, cross-geometry transfer or reactor safety certification. POWER/HEATFLUX scaling is unresolved and excluded. No industrial savings or critical heat-flux prediction is tested.

6. Evidence and use

run.py replays frozen equations without fitting. rules.json and native XML anchors make calculations auditable. Compact evidence retains all reserved runs and comparators. The evaluator scores alternative rules under the same information contract; admission and novelty remain scientific judgments. Targets are now exposed and rejected hypotheses remain in history.

Sources: [HWAT dataset, Zenodo 14627088](#); Le Corre et al. (2025), [source publication](#); [IAPWS saturation release](#). Drift flux follows Zuber and Findlay (1965), [doi:10.1115/1.3689137](#).